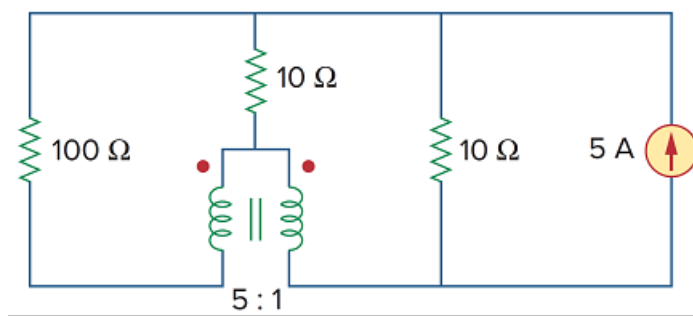


Problem

Find the power absorbed by the $100\text{-}\Omega$ resistor in the ideal transformer circuit of Fig. 13.121.

Figure 13.121



Step-by-step solution

Step 1 of 4

Refer to Figure 13.121 in the textbook for the ideal transformer circuit.

Convert the current source in the circuit into voltage source.

The current in the secondary side of the transformer is,

$$I_{s1} = 5 \text{ A}$$

Therefore,

$$\begin{aligned} V_{s1} &= I_{s1}R \\ &= (5 \text{ A})(10 \text{ }\Omega) \\ &= 50 \text{ V} \end{aligned}$$

Represent the loop currents in the circuit and draw the modified circuit.

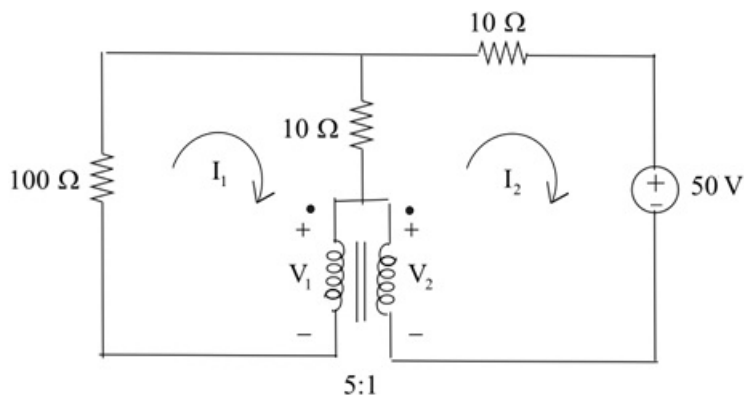


Figure 1

Step 2 of 4

Apply Kirchhoff's voltage law to the primary side of the transformer in the circuit.

$$100\mathbf{I}_1 + 10(\mathbf{I}_1 - \mathbf{I}_2) + \mathbf{V}_1 = 0$$

$$100\mathbf{I}_1 + 10\mathbf{I}_1 - 10\mathbf{I}_2 + \mathbf{V}_1 = 0$$

$$110\mathbf{I}_1 - 10\mathbf{I}_2 + \mathbf{V}_1 = 0$$

Apply Kirchhoff's voltage law to the secondary side of the transformer.

$$10\mathbf{I}_2 + 50 - \mathbf{V}_2 + 10(\mathbf{I}_2 - \mathbf{I}_1) = 0$$

$$10\mathbf{I}_2 + 50 - \mathbf{V}_2 + 10\mathbf{I}_2 - 10\mathbf{I}_1 = 0$$

$$20\mathbf{I}_2 - 10\mathbf{I}_1 - \mathbf{V}_2 = -50$$

Consider the following expression for the voltage ratio of the transformer:

$$\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{N_2}{N_1}$$

Substitute 1 for N_2 and 5 for N_1 .

$$\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{1}{5}$$

$$\mathbf{V}_2 = \frac{\mathbf{V}_1}{5}$$

Consider the following expression for the current ratio of the transformer:

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_2}{N_1}$$

Substitute 1 for N_2 and 5 for N_1 .

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{1}{5}$$

$$\mathbf{I}_2 = 5\mathbf{I}_1$$

Step 3 of 4

Substitute the expressions of $\mathbf{V}_2$ and $\mathbf{I}_2$ in the second KVL equation.

$$20\mathbf{I}_2 - 10\mathbf{I}_1 - \mathbf{V}_2 = -50$$

$$20(5\mathbf{I}_1) - 10\mathbf{I}_1 - \frac{\mathbf{V}_1}{5} = -50$$

$$100\mathbf{I}_1 - 10\mathbf{I}_1 - \frac{\mathbf{V}_1}{5} = -50$$

$$90\mathbf{I}_1 - \frac{\mathbf{V}_1}{5} = -50$$

$$450\mathbf{I}_1 - \mathbf{V}_1 = -250$$

$$\mathbf{V}_1 = 450\mathbf{I}_1 + 250$$

Substitute the expression of $\mathbf{I}_2$ in the first KVL equation.

$$110\mathbf{I}_1 - 10\mathbf{I}_2 + \mathbf{V}_1 = 0$$

$$110\mathbf{I}_1 - 10(5\mathbf{I}_1) + \mathbf{V}_1 = 0$$

$$110\mathbf{I}_1 - 50\mathbf{I}_1 + \mathbf{V}_1 = 0$$

$$60\mathbf{I}_1 + \mathbf{V}_1 = 0$$

Substitute the expression of $\mathbf{V}_1$ in the equation.

$$60\mathbf{I}_1 + (450\mathbf{I}_1 + 250) = 0$$

$$60\mathbf{I}_1 + 450\mathbf{I}_1 + 250 = 0$$

$$510\mathbf{I}_1 = -250$$

$$\begin{aligned}\mathbf{I}_1 &= \frac{-250}{510} \\ &= -0.49 \text{ A}\end{aligned}$$

Step 4 of 4

Calculate the power absorbed by the $100\ \Omega$ resistor in the primary side of the transformer in the circuit.

$$P_{100\Omega} = 100\mathbf{I_1^2}$$

Substitute $-0.49\ \text{A}$ for $\mathbf{I_1}$.

$$\begin{aligned} P_{100\Omega} &= 100(-0.49)^2 \\ &= 100(0.24) \\ &= 24\ \text{W} \end{aligned}$$

Thus, the power absorbed by $100\ \Omega$ resistor is 24 W.